

modulo4

July 1, 2026

1 Módulo 4: Modelos Avanzados de Pronóstico

Data Analytics con Pronósticos a través de Python

Duración: 4 horas

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import mean_absolute_error, mean_squared_error
import warnings
warnings.filterwarnings("ignore")

%matplotlib inline
plt.style.use('seaborn-v0_8')
```

1.1 1. Preparación de Datos

```
[3]: # Dataset de ejemplo
np.random.seed(42)
dates = pd.date_range('2023-01-01', periods=500, freq='D')
df = pd.DataFrame({
    'fecha': dates,
    'ventas': 1000 + np.sin(np.arange(500)*0.1)*300 + np.random.normal(0, 100, 500) + np.arange(500)*1.5
})
df.set_index('fecha', inplace=True)
df = df.asfreq('D')
df.head()
```

```
[3]:          ventas
fecha
2023-01-01  1049.671415
2023-01-02  1017.623595
2023-01-03  1127.369653
2023-01-04  1245.459048
2023-01-05  1099.410165
```

1.2 2. Prophet (Facebook)

```
[5]: pip install prophet tensorflow
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting prophet
  Downloading prophet-1.3.0-py3-none-manylinux2014_x86_64.whl.metadata (3.5 kB)
Requirement already satisfied: tensorflow in /home/fab/.local/lib/python3.10/site-packages (2.18.0)
Collecting cmdstanpy>=1.0.4 (from prophet)
  Downloading cmdstanpy-1.3.0-py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: numpy>=1.15.4 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (2.2.6)
Requirement already satisfied: matplotlib>=2.0.0 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (3.10.8)
Requirement already satisfied: pandas>=1.0.4 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (2.2.3)
Collecting holidays<1,>=0.25 (from prophet)
  Downloading holidays-0.99-py3-none-any.whl.metadata (53 kB)
Requirement already satisfied: tqdm>=4.36.1 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (4.67.1)
Collecting importlib_resources (from prophet)
  Downloading importlib_resources-7.1.0-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: python-dateutil<3,>=2.9.0.post0 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (1.17.0)
Requirement already satisfied: absl-py>=1.0.0 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (2.1.0)
Requirement already satisfied: astunparse>=1.6.0 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (25.2.10)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (0.6.0)
Requirement already satisfied: google-pasta>=0.1.1 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in /home/fab/.local/lib/python3.10/site-packages (from prophet) (3.4.0)
Requirement already satisfied: packaging in /home/fab/.local/lib/python3.10/site-packages (from prophet) (25.0)
Requirement already satisfied:
protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3
```

```

in /home/fab/.local/lib/python3.10/site-packages (from tensorflow) (5.29.3)
Requirement already satisfied: requests<3,>=2.21.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (2.32.3)
Requirement already satisfied: setuptools in /usr/lib/python3/dist-packages
(from tensorflow) (59.6.0)
Requirement already satisfied: termcolor>=1.1.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: typing-extensions>=3.6.6 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (1.17.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (1.70.0)
Requirement already satisfied: tensorboard<2.19,>=2.18 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (2.18.0)
Requirement already satisfied: keras>=3.5.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (3.8.0)
Collecting numpy>=1.15.4 (from prophet)
    Using cached
numpy-2.0.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(60 kB)
Requirement already satisfied: h5py>=3.11.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (3.13.0)
Requirement already satisfied: ml-dtypes<0.5.0,>=0.4.0 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (0.4.1)
Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in
/home/fab/.local/lib/python3.10/site-packages (from tensorflow) (0.37.1)
Requirement already satisfied: charset-normalizer<4,>=2 in
/home/fab/.local/lib/python3.10/site-packages (from
requests<3,>=2.21.0->tensorflow) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/lib/python3/dist-packages
(from requests<3,>=2.21.0->tensorflow) (3.3)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/lib/python3/dist-
packages (from requests<3,>=2.21.0->tensorflow) (1.26.5)
Requirement already satisfied: certifi>=2017.4.17 in /usr/lib/python3/dist-
packages (from requests<3,>=2.21.0->tensorflow) (2020.6.20)
Requirement already satisfied: markdown>=2.6.8 in
/home/fab/.local/lib/python3.10/site-packages (from
tensorboard<2.19,>=2.18->tensorflow) (3.7)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
/home/fab/.local/lib/python3.10/site-packages (from
tensorboard<2.19,>=2.18->tensorflow) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in
/home/fab/.local/lib/python3.10/site-packages (from
tensorboard<2.19,>=2.18->tensorflow) (3.0.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/lib/python3/dist-
packages (from astunparse>=1.6.0->tensorflow) (0.37.1)
Collecting stanio<2.0.0,>=0.4.0 (from cmdstanpy>=1.0.4->prophet)

```

Downloading stanio-0.5.1-py3-none-any.whl.metadata (1.6 kB)
 Requirement already satisfied: rich in /home/fab/.local/lib/python3.10/site-packages (from keras>=3.5.0->tensorflow) (13.9.4)
 Requirement already satisfied: namex in /home/fab/.local/lib/python3.10/site-packages (from keras>=3.5.0->tensorflow) (0.0.8)
 Requirement already satisfied: optree in /home/fab/.local/lib/python3.10/site-packages (from keras>=3.5.0->tensorflow) (0.14.0)
 Requirement already satisfied: contourpy>=1.0.1 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (1.3.2)
 Requirement already satisfied: cycycler>=0.10 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (4.61.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (1.4.9)
 Requirement already satisfied: pillow>=8 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (12.1.0)
 Requirement already satisfied: pyparsing>=3 in /home/fab/.local/lib/python3.10/site-packages (from matplotlib>=2.0.0->prophet) (3.3.1)
 Requirement already satisfied: pytz>=2020.1 in /home/fab/.local/lib/python3.10/site-packages (from pandas>=1.0.4->prophet) (2025.1)
 Requirement already satisfied: tzdata>=2022.7 in /home/fab/.local/lib/python3.10/site-packages (from pandas>=1.0.4->prophet) (2024.2)
 Requirement already satisfied: MarkupSafe>=2.1.1 in /home/fab/.local/lib/python3.10/site-packages (from werkzeug>=1.0.1->tensorboard<2.19,>=2.18->tensorflow) (2.1.5)
 Requirement already satisfied: markdown-it-py>=2.2.0 in /home/fab/.local/lib/python3.10/site-packages (from rich->keras>=3.5.0->tensorflow) (3.0.0)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /home/fab/.local/lib/python3.10/site-packages (from rich->keras>=3.5.0->tensorflow) (2.18.0)
 Requirement already satisfied: mdurl~=0.1 in /home/fab/.local/lib/python3.10/site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow) (0.1.2)
 Downloading prophet-1.3.0-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (12.1 MB)

12.1/12.1 MB 9.3 MB/s 0:00:01m 8.4 MB/s eta
 0:00:01

Downloading holidays-0.99-py3-none-any.whl (1.5 MB)

1.5/1.5 MB 18.9 MB/s 0:00:00

Using cached

numpy-2.0.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (19.5 MB)

Downloading cmdstanpy-1.3.0-py3-none-any.whl (99 kB)

Downloading stanio-0.5.1-py3-none-any.whl (8.1 kB)

Downloading importlib_resources-7.1.0-py3-none-any.whl (37 kB)

WARNING: Error parsing dependencies of python-apt: Invalid version:

'2.4.0-ubuntu4.1-zorin1'

Installing collected packages: numpy, importlib_resources, stanio, holidays, cmdstanpy, prophet

Attempting uninstall: numpy

Found existing installation: numpy 2.2.6

Uninstalling numpy-2.2.6:

Successfully uninstalled numpy-2.2.6

6/6 [prophet] 5/6 [prophet]]

ERROR: pip's dependency resolver does not currently take into

account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

gradio 5.35.0 requires pillow<12.0,>=8.0, but you have pillow 12.1.0 which is incompatible.

Successfully installed cmdstanpy-1.3.0 holidays-0.99

importlib_resources-7.1.0 numpy-2.0.2 prophet-1.3.0 stanio-0.5.1

[notice] A new release of pip is

available: 26.0.1 -> 26.1.2

[notice] To update, run:

python3 -m pip install --upgrade pip

Note: you may need to restart the kernel to use updated packages.

```
[6]: from prophet import Prophet

df_prophet = df.reset_index()[['fecha', 'ventas']].rename(columns={'fecha':
    ↳ 'ds', 'ventas': 'y'})

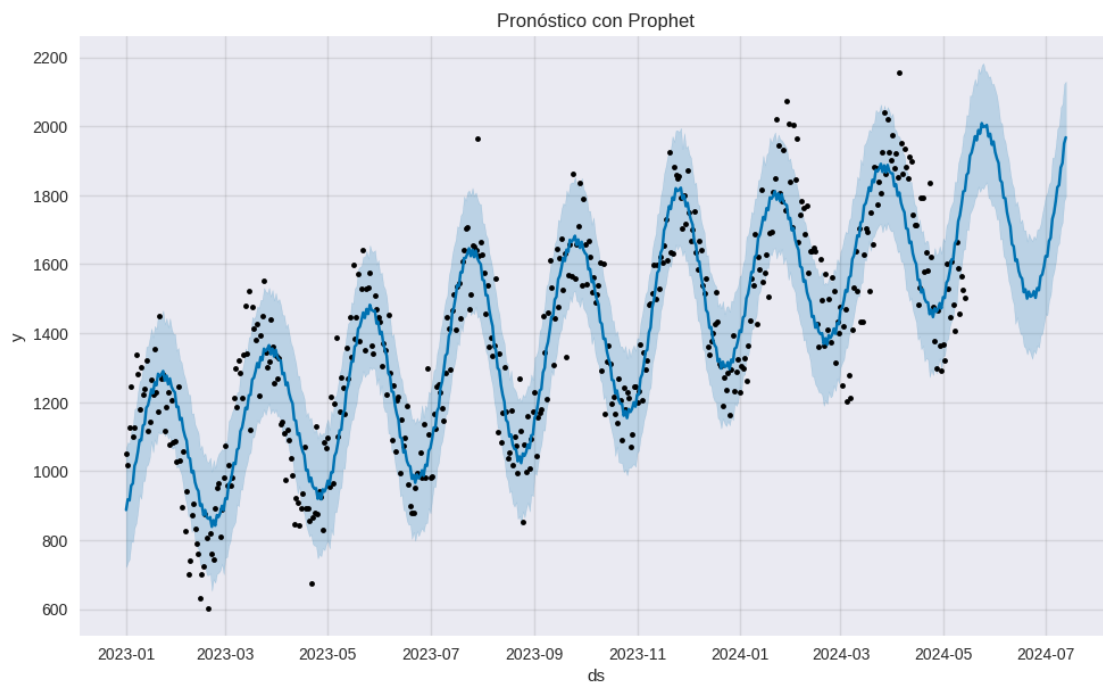
model_p = Prophet(yearly_seasonality=True, weekly_seasonality=True,
    ↳ daily_seasonality=False, seasonality_mode='additive')
model_p.fit(df_prophet)

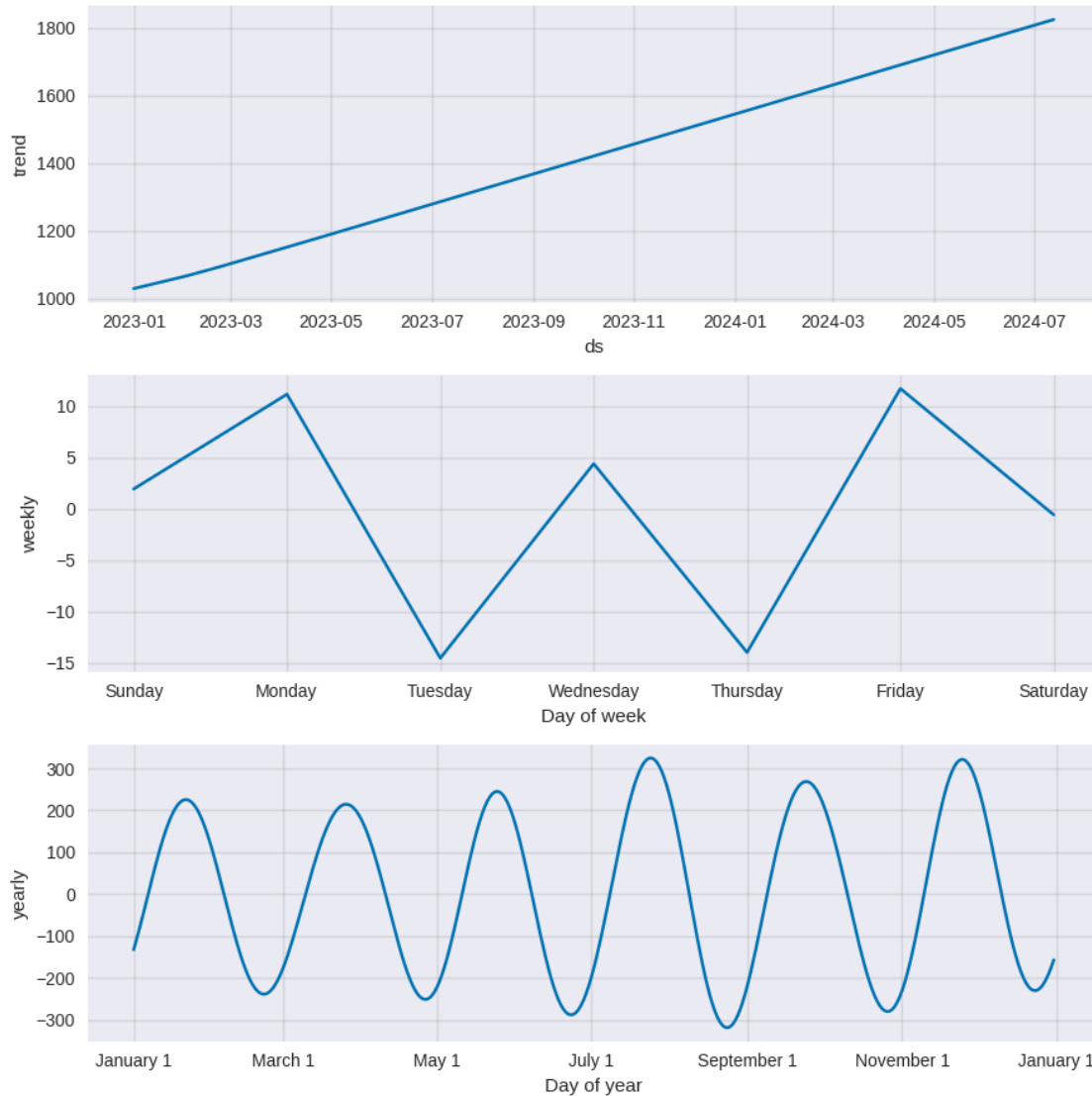
future = model_p.make_future_dataframe(periods=60)
forecast_p = model_p.predict(future)
```

```
model_p.plot(forecast_p)
plt.title('Pronóstico con Prophet')
plt.show()

model_p.plot_components(forecast_p)
plt.show()
```

```
12:34:31 - cmdstanpy - INFO - Chain [1] start processing
12:34:31 - cmdstanpy - INFO - Chain [1] done processing
```





1.3 3. Modelos de Machine Learning

```
[7]: # Crear features
df_ml = df.copy()
df_ml['mes'] = df_ml.index.month
df_ml['dia_semana'] = df_ml.index.dayofweek
df_ml['lag_1'] = df_ml['ventas'].shift(1)
df_ml['lag_7'] = df_ml['ventas'].shift(7)
df_ml.dropna(inplace=True)

X = df_ml[['mes', 'dia_semana', 'lag_1', 'lag_7']]
y = df_ml['ventas']
```

```
[11]: pip install xgboost
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting xgboost
  Downloading xgboost-3.2.0-py3-none-manylinux_2_28_x86_64.whl.metadata (2.1 kB)
Requirement already satisfied: numpy in /home/fab/.local/lib/python3.10/site-packages (from xgboost) (2.0.2)
Requirement already satisfied: nvidia-nccl-cu12 in /home/fab/.local/lib/python3.10/site-packages (from xgboost) (2.27.5)
Requirement already satisfied: scipy in /home/fab/.local/lib/python3.10/site-packages (from xgboost) (1.15.1)
Downloading xgboost-3.2.0-py3-none-manylinux_2_28_x86_64.whl (131.7 MB)

131.7/131.7 MB 7.6 MB/s 0:00:17 eta
0:00:01[36m0:00:01
WARNING: Error parsing dependencies of python-apt: Invalid version:
'2.4.0-ubuntu4.1-zorin1'

Installing collected packages: xgboost
Successfully installed xgboost-3.2.0

[notice] A new release of pip is
available: 26.0.1 -> 26.1.2
[notice] To update, run:
python3 -m pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

```
[12]: from sklearn.model_selection import TimeSeriesSplit
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor

# Random Forest
rf = RandomForestRegressor(n_estimators=200, random_state=42)
rf.fit(X, y)

# XGBoost
xgb = XGBRegressor(n_estimators=200, learning_rate=0.05, max_depth=6,
↳ random_state=42)
xgb.fit(X, y)

print(" Modelos de Machine Learning entrenados")
```

Modelos de Machine Learning entrenados

1.4 4. Evaluación de Modelos

```
[13]: def evaluar(y_true, y_pred, nombre):  
    mae = mean_absolute_error(y_true, y_pred)  
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))  
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100  
    print(f"{nombre:15} → MAE: {mae:6.2f} | RMSE: {rmse:6.2f} | MAPE: {mape:5.2f}%")  
  
    # Evaluación (ejemplo con datos de prueba)  
    print("Evaluación de Modelos:")  
    evaluar(y[-60:], rf.predict(X[-60:]), "Random Forest")  
    evaluar(y[-60:], xgb.predict(X[-60:]), "XGBoost")
```

Evaluación de Modelos:

Random Forest	→ MAE:	42.91	RMSE:	52.49	MAPE:	2.59%
XGBoost	→ MAE:	36.54	RMSE:	50.48	MAPE:	2.19%

¡Fin del Módulo 4!

Tarea Final: Realiza una comparativa completa entre Prophet, ARIMA, Random Forest y XGBoost usando el mismo dataset.